

Lecture 4 – Preserving Privacy

Disclaimer: These are my personal study notes created for learning and revision while taking Harvard CS50's Introduction to Cybersecurity. They are not official course materials.

Preserving Privacy

Privacy is about understanding what information is shared while using the internet and learning how to control or reduce unnecessary sharing.

Web Browsing History

Your browsing history is useful because it helps you revisit previously visited websites. However, it can also become a privacy risk if someone gains access to it.

You can clear your browser history to remove local records. However, doing so may also log you out of websites and online services.

Even if you clear your browser history, websites may still keep records of your activity through **server logs**.

Example of a server log:

```
log_format combined '$remote_addr - $remote_user [$time_local] '
                    '"$request" $status $body_bytes_sent '
                    '"$http_referer" "$http_user_agent"';
```

This server log may contain information such as:

- **\$remote_addr** – User's IP Address
- **\$remote_user** – User Information []
- **\$time_local** – Local Date and Time
- **\$request** – Requested Resource
- **\$status** – HTTP Status Code
- **\$body_bytes_sent** – Amount of Data Sent
- **\$http_referer** – Referring Website
- **\$http_user_agent** – Browser or Device Information

This shows that information such as your IP address, request details, browser information, and other data are exchanged between your browser and the server.

Therefore, clearing your browser history **does not remove the records stored on the server**.

NOTE:

- If server logs are leaked, they may expose sensitive user information, creating both privacy and security risks.
- Server log leaks can be minimized by encrypting logs, restricting access, storing only necessary data, and securely managing or deleting old logs.

HTTP Headers

What are HTTP Headers?

HTTP Headers are **key-value pairs** exchanged between the browser and the server. They provide additional information whenever a request or response is sent.

Example

Consider the following HTML:

```
<a href="https://example.com">cats</a>
```

This creates a hyperlink named **cats** that redirects the user to **example.com**.

Suppose the user first searches "**cats**" on Google and then clicks this link.

By default, the browser automatically sends the following HTTP header:

```
Referer: https://www.google.com/search?q=cats
```

The **Referer** header tells the destination website **which webpage the user came from**.

In this example, **example.com** can see that the user came from Google and searched for "**cats**" before visiting the website.

This may reveal more information than the user intends to share.

Sharing Only the Website Origin

Instead of sending the complete URL, the browser can send only the website's origin.

Example:

Referer: <https://www.google.com/>

Now the destination website only knows that the visitor came from **Google**, but it cannot see the exact search query.

This can be achieved by adding the following meta tag:

```
<meta name="referrer" content="origin">
```

The **origin** value instructs the browser to share only the website's origin.

Preventing Referrer Information

If a website does not want to share any referrer information, it can use:

```
<meta name="referrer" content="none">
```

The **none** value tells the browser **not to send any Referer header**, providing better privacy.

Key Points to Remember

- HTTP Headers are key-value pairs exchanged between the browser and the server.
- The **Referer** header tells a website where the visitor came from.
- By default, the browser may share the complete referring URL.
- `content="origin"` shares only the website's origin.
- `content="none"` prevents any referrer information from being shared, improving user privacy.

Who does what?

- **Developer/Website (Server):** Decides the referrer policy (`origin`, `none`, etc.).
- **Browser:** Reads that policy and follows it when sending requests.
- **User:** Normally does nothing (unless they use browser privacy settings or extensions to block referrers).

One-line note:

The website developer sets the Referrer Policy, and the browser enforces it when sending HTTP requests.

Fingerprinting

What is Fingerprinting?

Fingerprinting is a technique used by websites or third parties to **identify and distinguish users** based on the information their browser and device automatically reveal.

Even if you limit the amount of information your browser shares, enough details can still be collected to uniquely identify you.

Browser Information

Every browser shares some information about the user's device while communicating with a web server.

Different browsers may share **more or less information** depending on their settings and privacy features.

However, regardless of the browser used, **web servers can still record user activity**.

User-Agent Header

One common source of information is the **User-Agent** request header.

Example:

```
Mozilla/5.0 (Linux; {Android Version}; {Build Tag etc.})  
AppleWebKit/{WebKit Rev} (KHTML, like Gecko)  
Chrome/{Chrome Rev} Mobile Safari/{WebKit Rev}
```

The **User-Agent** header may reveal information such as:

- Browser Name
 - Browser Version
 - Operating System
 - Device Type
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Other Information Used for Fingerprinting

In addition to the User-Agent header, web servers may also collect information such as:

- IP Address
 - Screen Resolution
 - Installed Browser Extensions
 - Installed Fonts
 - Other Device Information
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Why is Fingerprinting a Privacy Concern?

When all of this information is collected together over time, it can create a **unique fingerprint** that makes a user easier to identify and track across websites, even without relying on cookies.

NOTE:

- **Fingerprinting can be reduced by limiting the information shared by the browser, using privacy-focused browsers, minimizing extensions, blocking trackers, and hiding the IP address with a VPN.**
- **Remember:** Unlike cookies, **fingerprinting is difficult to completely block** because it relies on many small pieces of information that your browser naturally exposes.

Key Points to Remember

- Fingerprinting identifies users using information shared by their browser and device.
- Different browsers reveal different amounts of information.
- The **User-Agent** header provides details about the browser, operating system, and device.
- Websites may also collect the IP address, screen resolution, installed fonts, extensions, and other device information.
- Combining these details over time makes a user more easily identifiable and trackable.

Session Cookies

Definition

- Session cookies are temporary cookies used to identify a user during a browsing session.
- They act like a temporary ID assigned by the server.

Example

HTTP/3 200

Set-Cookie: session=1234abcd

Explanation

- Set-Cookie creates a cookie named **session**.
 - 1234abcd is a unique session ID for the user.
 - Each user receives a different session ID.
 - Session cookies expire after a certain time or when the session ends.
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Tracking Cookies

Definition

- Tracking cookies monitor a user's activity across websites.
- They are commonly used for analytics, advertising, and personalization.

Example

Set-Cookie: `_ga=GA1.2.0123456789.0; max-age=63072000`

Explanation

- `_ga` is a Google Analytics cookie.
 - `max-age` specifies how long the cookie is stored.
 - In this example, the cookie remains active for **2 years**.
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Tracking Parameters

Definition

- Tracking parameters are values added to a URL to record user interactions.
- They are visible in the browser's address bar.

Example

`https://example.com/ad_engagement?click_id=YmVhODI1MmZmNGU4&campaign_id=23`

Explanation

- `click_id` uniquely identifies a user's click.
- `campaign_id` identifies the advertising campaign.

Privacy-Friendly Example

`https://example.com/ad_engagement?campaign_id=23`

- Only the campaign is tracked.
 - No unique user identifier is included.
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Third-Party Cookies

Definition

- Third-party cookies are created by a website different from the one the user is visiting.
- They help track users across multiple websites.

Example Request

GET /ad.gif HTTP/3
Host: example.com
Referer: https://harvard.edu/

Server Response

HTTP/3 200
Set-Cookie: id=1234abcd; max-age=31536000

Later Request

GET /ad.gif HTTP/3
Cookie: id=1234abcd
Host: example.com
Referer: https://yale.edu/

Explanation

- The same cookie is sent whenever content is loaded from example.com.
- This allows the third-party server to recognize the same user across different websites(Harvard & Yale).
- Third-party cookies are mainly used for advertising and analytics.

Easy Memory Trick :

- Think of **Google Ads**.
- Many websites show Google advertisements.
- Google can recognize your browser on all those websites using the same cookie.

NOTE: A third-party cookie is a cookie created by a different website than the one the user is visiting, allowing the third party to recognize and track the user across multiple websites.

NOTE:

- **Session cookies are different for each website.** Each website creates its own unique session ID, so `harvard.edu` and `yale.edu` will have different session cookies.
 - **Exception: A third-party cookie** (from the same third-party domain, like `example.com`) can be the same across multiple websites that load content from that third party.
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Private Browsing

Definition

- Private browsing limits the storage of browsing data after the session ends.

Key Points

- When you open a private window, **old cookies and browsing history are not used**, but websites can still create **new cookies** and track you during that private session.
 - When you close the private window, **those new cookies, history, and other browsing data are deleted from your device**, but the websites you visited could still have recorded your activity while you were using them.
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Supercookies

- Your **Internet Service Provider (ISP)** may add special tracking identifiers (**supercookies**) to your internet traffic without your knowledge.
 - **Some ISPs give you an option to turn off this tracking**, so they stop adding supercookies to your internet traffic.
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Key Points to Remember

- **Session Cookies** temporarily identify a user during a browsing session.
- **Tracking Cookies** monitor user activity over time.
- **Tracking Parameters** are visible values in a URL used to track clicks or campaigns.
- **Third-Party Cookies** follow users across different websites.
- **Private Browsing** removes local browsing data after the session but does not stop tracking during the session.
- **Supercookies** are network-level tracking identifiers that may be added by an ISP.

DNS (Domain Name System)

Definition

- DNS converts a website name (like **harvard.edu**) into its corresponding IP address.
- DNS converts a website name into the correct IP address, allowing the browser to connect to the appropriate server (or the nearest server for large websites).

Key Points

- **Normal DNS requests are not encrypted.**
 - **Your ISP and DNS provider can see which websites you are trying to visit.**
 - **DNS over HTTPS (DoH) and DNS over TLS (DoT) encrypt DNS requests so others cannot easily see which website you are trying to visit. (This improves privacy by hiding your DNS queries from people monitoring the network).**
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Virtual Private Network (VPN)

Definition

- A VPN creates an encrypted connection between your device and a VPN server.
- Websites see the **VPN server's IP address** instead of your real IP address.

Key Points

- Hides your real IP address.
 - Encrypts internet traffic between your device and the VPN server.
 - Improves online privacy.
 - Does **not** protect your device if it is infected with malware.
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Tor

Definition

Tor Browser is a web browser that uses the Tor network to provide anonymous and private browsing.

- Tor is software that routes your internet traffic through multiple encrypted servers (nodes).
- This makes it much harder to identify your real location or identity.

Key Points

- Traffic passes through several encrypted nodes.
 - Tor stores very little information about your browsing activity.
 - Provides a higher level of privacy than normal browsing.
 - It does **not** guarantee complete anonymity(Privacy) in every situation.
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Permissions

Definition

- Applications and operating systems request permission to access features on your device, such as location.

Key Points

- Location permission allows apps to know your geographic location.
 - Apps like Apple Maps and Google Maps can track your location if permission is granted.
 - Only grant permissions when they are necessary.
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Key Points to Remember

- **DNS** converts website names into IP addresses.
- **DoH** and **DoT** encrypt DNS requests for better privacy.
- **VPN** hides your IP address and encrypts your internet connection.
- **Tor** improves privacy by routing traffic through multiple encrypted servers.
- **Permissions** control what information apps can access on your device.